Steel deck design review

01 / SUMMARY OF CHECKS

Check	Demand	Limit	Ratio	Comparison
Composite flexure	9.6932	21.59	0.449	Passed
Composite vertical shear	12.9243	50.41	0.2564	Passed
Construction flexure	6.1372	--	--	NOT CHECKED
Construction shear	8.1829	--	--	NOT CHECKED
Construction bearing	8.1829	--	--	NOT CHECKED
Construction dead-load deflection	20.7199	16.6667	1.2432	Fail
Composite live deflection	--	8.3333	--	NOT CHECKED
Composite total deflection	--	12.5	--	NOT CHECKED

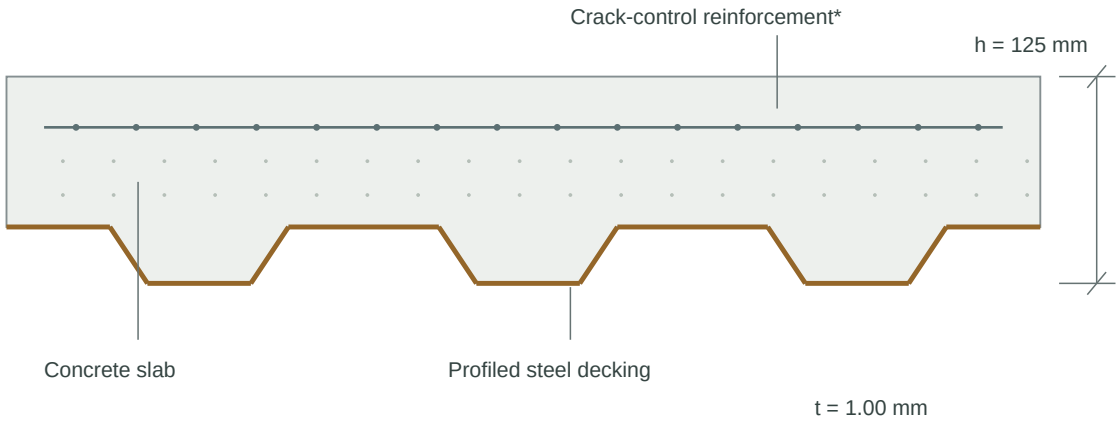
Units: moment kN.m/m; shear/bearing kN/m; deflection mm. Passed applies only to that comparison. -- means unavailable. Ratios use unrounded values; limit = 1.0000.

Project / selection	Record
Project	Proposed steel deck floor
Manufacturer / profile	Architecks Metal Systems Inc. / DN Steel Deck Four
Section / analysis spans	t = 1.00 mm; h = 125 mm; L = 3 m; Lc = 3 m

PRELIMINARY / INCOMPLETE DESIGN - NOT FOR CONSTRUCTION

Fail - evaluated limit exceeded

TYPICAL COMPOSITE SLAB CROSS-SECTION



Typical transverse cross-section - schematic, not to scale. Rib geometry is illustrative. *Reinforcement size, spacing and cover require separate design. Span runs perpendicular to this section.

Design inputs & catalog properties

02 / SELECTED PRODUCT - AUTOMATIC CATALOG LOOKUP

Architecks Metal Systems Inc. / DN Steel Deck Four. Supplied Architecks catalog, PDF page 6. Exact listed thickness and overall slab depth; no interpolation.

Catalog property	Value	Unit
Steel deck mass	10	kg/m2
Concrete volume	0.102	m3/m2
Steel deck inertia, Is	648000	mm4 per m strip
Composite average inertia, Iave	12066492	mm4 per m strip
Composite flexure resistance, catalog oM	21.59	kN.m/m
Composite shear resistance, catalog oV	50.41	kN/m

PROJECT INPUTS & CRITERIA

Parameter	Value	Unit / basis
Prepared by	Not specified	Project record
Completed slab analysis span, L	3	m
Construction analysis span, Lc	3	m
Superimposed dead load	1	kPa
Occupancy live load	2.76	kPa
Uniform construction live load	1.5	kPa
Wet / dry concrete unit weight	24 / 23.55	kN/m3
Steel modulus, Es	200000	N/mm2
Concrete modulus, Ec	23000	N/mm2
Iave transformation basis	unconfirmed	Supplier confirmation
Deflection limits: construction / live / total	180 / 360 / 240	Span / denominator
Construction moment resistance	Not provided	kN.m/m, factored
Construction shear resistance	Not provided	kN/m, factored
Construction bearing resistance	Not provided	kN/m, factored

Additional resistance source: Not supplied. Optional construction resistances remain NOT CHECKED.

Catalog properties are populated from the selected product. Project loads, concrete properties and deflection criteria are user assumptions. Catalog resistance and stiffness bases require supplier verification.

Manual calculation trail

03 / FORMULAS, SUBSTITUTIONS & RESULTS

Simply supported, one-way, 1.000 m strip. 1 kPa x 1 m = 1 kN/m = 1 N/mm. Spans are converted from m to mm for deflection. Display values are rounded; calculations retain full precision.

01 Deck self-weight

wdeck = mass x g / 1000
Substitute: 10 x 9.80665 / 1000

Result = 0.0981 kPa

Catalog mass in kg/m2; standard gravity in m/s2.

02 Concrete self-weight

wwet = V x gammawet; wdry = V x gammadry
Substitute: V = 0.102 m3/m2; wet: 0.102 x 24; dry: 0.102 x 23.55

Result = 2.4021 kPa (dry)

Wet weight = 2.448 kPa. Profile concrete volume, not overall depth times unit area.

03 Permanent dead load

D = wdry + wdeck + SDL
Substitute: 2.4021 + 0.0981 + 1

Result = 3.5002 kPa

SDL includes user-entered finishes, ceiling and permanent partitions.

04 Composite design load

wu = max(1.4D, 1.2D + 1.6L)
Substitute: max(4.9002, 1.2 x 3.5002 + 1.6 x 2.76)

Result = 8.6162 kN/m

1 m strip; gravity D/L combinations only.

05 Composite sagging moment

Mu = wu L^2 / 8
Substitute: 8.6162 x 3^2 / 8

Result = 9.6932 kN.m/m

Simply supported uniform load.

06 Composite support shear

Vu = wu L / 2
Substitute: 8.6162 x 3 / 2

Result = 12.9243 kN/m

At support; no critical-section reduction.

Manual calculation trail / continued

07 Construction design load

wc = max(1.4Dc, 1.2Dc + 1.6CL)
Substitute: Dc = 2.448 + 0.0981 = 2.5461; max(3.5645, 5.4553)
Result = 5.4553 kN/m

Uniform-load screening only; CL is a project input.

08 Construction moment

Mc = wc Lc^2 / 8
Substitute: 5.4553 x 3^2 / 8
Result = 6.1372 kN.m/m

Deck alone before concrete gains strength.

09 Construction support shear

Vc = wc Lc / 2
Substitute: 5.4553 x 3 / 2
Result = 8.1829 kN/m

Also the reaction for the entered bearing resistance comparison.

10 Construction dead-load deflection

delta = 5 Dc (1000Lc)^4 / (384 Es Is)
Substitute: 5 x 2.5461 x (3 x 1000)^4 / (384 x 200000 x 648000)
Result = 20.7199 mm

Dc in N/mm for 1 m strip; effective stiffness requires verification. Excludes construction live load.

11 Composite flexure (catalog) comparison

utilization = demand / resistance or limit
Substitute: 9.6932 / 21.59
Result = 0.449

Passed; Catalog p.6, oM; supplier basis unconfirmed. Limit <= 1.0000; comparison uses unrounded numbers.

12 Composite vertical shear (catalog) comparison

utilization = demand / resistance or limit
Substitute: 12.9243 / 50.41
Result = 0.2564

Passed; Catalog p.6, oV; bond is separate. Limit <= 1.0000; comparison uses unrounded numbers.

13 Construction dead-load deflection comparison

utilization = demand / resistance or limit
Substitute: 20.7199 / 16.6667
Result = 1.2432

Fail; Project criterion Lc/180. Limit <= 1.0000; comparison uses unrounded numbers.

Design basis & review record

04 / ASSUMPTIONS & OUTSTANDING VERIFICATION

01. Scope: preliminary one-way, simply supported, 1 m strip under uniform static gravity loads. L and Lc are analysis spans between reaction lines. Props require separate design.

02. Product: Architecks DN Steel Deck Four, exact catalog rows only. No interpolation. Catalog oM and oV are provisional design resistances; do not interchange oM with the separate OM column. Confirm concrete strength, resistance basis, applicable spans and support conditions with the supplier.

03. Construction: uniform load screening only. Construction point loads, positioning, impact, erection sequence, wet-concrete accumulation and ponding are not evaluated. Entered resistances must be factored, for this section and support detail.

04. Deflection: instantaneous elastic calculation only. Match Iave with the modulus of its confirmed reference material (Es or Ec); otherwise composite deflection is NOT CHECKED. delta = 5wL^4/(384 Eref Iave). Construction deflection uses wet concrete plus deck only.

05. Serviceability: deflection denominators are project criteria. No creep, shrinkage, cracking recalculation, staged stress, load history or long-term deflection is included.

06. Outstanding strength checks: longitudinal shear/bond and incomplete shear connection; effective cold-formed section/local buckling; web crippling, actual bearing length, fasteners and end anchorage.

07. Outstanding detailing/system checks: shrinkage, temperature, negative-moment and fire reinforcement; diaphragm, openings, vibration, concentrated loads, seismic and fire design. Figure reinforcement is illustrative, not a designed detail.

08. Approval: verify occupancy loads, partitions, material strengths, site conditions and governing Philippine code edition. Passed is not an overall compliance or safe-for-construction conclusion. Engineer review and supplier confirmation remain required.

SOURCE REFERENCES

[1] Architecks Metal Systems Inc., supplied catalog, PDF page 6: DECK FOUR, Deck Properties and Composite Properties. Capacities and Iave are supplier values, not independently derived resistances.

[2] Supplied one-page Analysis and Design of Composite Slab (Steel Deck Slab): worked example only; different geometry and concrete density. Not a design standard or a validated benchmark.

[3] Philippine basis for engineer confirmation: NSCP C101-15, Volume I, 7th edition (2015), gravity strength combinations 1.4D and 1.2D + 1.6L. Only D and L are evaluated here. Confirm governing edition and local approval requirements.

[4] ANSI/SDI C-2017: composite steel floor deck-slabs reference. ANSI/SDI SD-2022 supersedes the separate 2017 standards; adopting it requires project-specific review. https://sdi.org/codes-standards/sdi-standards/

REVIEW RECORD

Prepared by: Not specified | Generated: 20/10/2026, 12:11:21 SGT

Supplier confirmation / date: _____

Engineer review / registration / signature / date: _____